12TH CHEMISTRY · CHAPTER 31 · CHAPTER TEST

Materials and Energy

Chapter-wise test: 6 MCQs, 5 of 6 short questions, 1 long question

Class: 12th (HSSC Part II)	**Subject:** Chemistry	**Total Marks:** 26
Chapter: 31	**Time Allowed:** 45 minutes	**Name:** _____

OBJECTIVE PART (6 × 1 = 6)

Q.1 Choose the correct answer.

1. Heating a carbonate ore in limited air is:

(a) Roasting (b) Alloying
(c) Refining (d) Calcination

2. Brass is an alloy of copper and:

(a) Tin (b) Zinc
(c) Iron (d) Carbon

3. The most common semiconductor is:

(a) Silicon (b) Copper
(c) Alumina (d) Carbon

4. The catalyst in catalytic cracking is:

(a) Platinum (b) Nickel
(c) Zeolite (d) Iron

5. Specific energy is measured in:

(a) MJ/L (b) MJ/kg
(c) kg/L (d) MJ/s

6. Fission of uranium-235 gives barium and:

(a) Radon (b) Helium
(c) Krypton (d) Xenon

SUBJECTIVE PART — SHORT QUESTIONS (5 × 2 = 10)

Q.2 Write short answers to any FIVE (5) questions.

i. Differentiate between roasting and calcination.
ii. What is a composite? Give one example.
iii. State two challenges of recycling.
iv. Give two applications of X-ray crystallography.
v. Differentiate energy density and specific energy.
vi. Why are batteries used in solar energy systems?

LONG QUESTION (2 × 5 = 10)

Q.3 Answer both parts.

(a) Describe the steps in the extraction of metals from ores.
(b) Compare thermal cracking and catalytic cracking.

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